

1 Question (12 points)

Prove that the language $A = \{1^{n(n+1)/2} | n \geq 0\}$ is not regular.

2 Question (12 points)

Prove that the language $B = \{a^{n-m}b^nc^m \mid n \geq m \geq 0\}$ is not regular.

3 Question (16 points)

Let $\Sigma = \{a, b\}$. For each $k \geq 1$ and $m < k$, let C_k be the language consisting of all strings that contain m **a**'s consequently starting from exactly k places from the right end. Describe an DFA with 2^k states that recognizes C_k .